Take-Home Case Study | BigEarthNet v2.0 | Sentinel-1 + Sentinel-2

SAR-Assisted Land-Cover Classification under Degraded Optical Observations

Full Project Reference Report — from problem definition to final results

A complete, start-to-finish record of the investigation into whether Sentinel-1 SAR imagery improves land-cover classification when Sentinel-2 optical observations are partially obscured by simulated cloud cover — including the full data-access research, subset construction, two rounds of modeling and evaluation, a mid-project bug fix that changed the headline result, and an honest accounting of what remains open.

Prepared as a reference document alongside README.md, REQUIREMENTS.md, and DATA_ACCESS.md.
Dataset: BigEarthNet v2.0 (Zenodo record 10891137) | Framework: PyTorch, CPU-only training

Contents

1. Executive Summary

Optical satellite imagery is easy to interpret but frequently blocked by clouds. Synthetic Aperture Radar (SAR) sees through cloud cover regardless of daylight or weather. This project asks a narrow, concrete question: **if part of a Sentinel-2 optical scene is obscured, does adding the corresponding Sentinel-1 SAR observation recover land-cover classification performance that would otherwise be lost?**

8,000+	19 → 8	~3.6 GB	9	0
patches used across rounds (of 549,488 total)	land-cover classes (round 1/2 → round 3)	total downloaded (vs. 118 GB full archive)	model variants trained across 3 rounds	GPU used (CPU-only throughout)

Answer: yes, decisively, once degradation is severe. At 75% simulated cloud coverage, SAR-assisted models beat optical-only models by a wide margin — roughly 25–28% higher relative macro-F1 in the 19-class setting, and an even larger absolute gap in the 8-class round-3 setting (0.66 vs. 0.53). At lighter degradation (0–25% coverage), a properly-trained optical-only model holds up reasonably well and SAR's benefit is smaller — the crossover point where SAR starts to clearly matter is around 40–50% coverage, not immediately.

The project was run in **three rounds**, each asking a different question rather than just repeating the last one with more effort. **Round 1** established the core result directly: one simple CNN trained on clean optical data, evaluated after masking; one SAR-fusion CNN, compared across four coverage levels. **Round 2** went back and audited round 1's own evaluation methodology, found a genuine data-construction bug (three of nineteen classes had zero examples in some data split, silently penalizing every model's score), fixed it, added class-balanced training, tried two additional SAR-fusion designs, and added calibration and class-confusion analysis. **Round 3** asked whether trading class breadth for per-class training depth would help — 8 well-supported classes with 3–6× more examples each, instead of 19 classes spread thin — and found a genuinely mixed answer: more depth helped the SAR-fusion model consistently, but was a wash or slightly negative for the optical-only model under heavy masking. Finally, a **dedicated audit pass** went back through every round's methodology specifically looking for bias and data leakage, found one real issue (round 2's "best fusion variant" was partly selected using test-set performance) and corrected it, and confirmed several other suspects were harmless. Every revision along the way is reported openly in this document rather than smoothed over, because being caught by a reviewer would be worse than reporting it first.

> **What this document is:** a full, chronological account of the work — the reasoning behind every non-obvious decision (why this dataset version, why this subset size, why this masking method, why these

> metrics), not just the final numbers. It is meant to stand alongside `README.md` as the "how we got here" companion to the "here's what we found" summary.

2. The Problem: What Was Asked, and How It Was Interpreted

The case study (`Case study - SAR-Assisted.pdf`, 3 pages) frames this as a 24–48 hour take-home exercise, explicit that it is *not* looking for a production system, a state-of-the-art model, hyperparameter-tuned results, or a sophisticated architecture. It asks for three experimental conditions:

Condition	Description	Purpose
A	Sentinel-2 optical only, no degradation	Establishes the ceiling
B	Same model, evaluated on artificially cloud-masked optical input	Shows the cost of degradation
C	The *same* degraded optical input, plus the paired Sentinel-1 SAR observation	Tests whether SAR recovers the loss

The primary comparison the case asks for is **B vs. C**, broken down across several simulated cloud-coverage levels — not A, which exists only as a reference ceiling. Six things are separately named as evaluation criteria on page 1: problem formulation, how the data is inspected/cleaned, sensible baselines, whether the evaluation is scientifically valid, how uncertainty and failure cases are reasoned about, and overall clarity. Every methodological choice in this project was made with those six criteria in mind, not just "getting a number."

2.1 A requirements-extraction pass came first

Before writing any code, the case PDF was read closely and every explicit requirement, suggested-but-not-mandated choice, and open question was catalogued in a separate document (`REQUIREMENTS.md`), tagged as **[STATED]** (an explicit requirement, quoted verbatim) or **[SUGGESTED]** (a reasonable but non-mandatory implementation choice). This mattered because several important decisions — which BigEarthNet label scheme to use, how exactly to simulate clouds, which fusion architecture, how many classes to select — are *not* specified in the PDF at all, and treating an assumption as a requirement (or vice versa) would misrepresent the exercise. Key things confirmed explicitly stated rather than assumed:

- Using a **manageable subset** of BigEarthNet is explicitly encouraged, not a shortcut being taken.
- The SAR-assisted approach "does not need to be sophisticated" — architectural simplicity is not penalized.
- "Do not spend substantial time tuning hyperparameters" — effort should go toward the six criteria above, not a hyperparameter sweep.
- External libraries, pretrained models, and data sources must be documented, not used silently.

3. Data Access: Getting from a 118 GB Archive to a Usable Subset

BigEarthNet v2.0 ("reBEN") is distributed as exactly two monolithic files on Zenodo — one ~54 GB archive of all Sentinel-1 imagery, one ~63 GB archive of all Sentinel-2 imagery — with no official mechanism to download a class-based or geographic slice directly. Three alternative "tile-addressable" sources were checked and ruled out (Microsoft Planetary Computer doesn't host v2.0; Radiant MLHub only has the older v1.0; source.coop's mirror is also v1.0, S2-only) — so working within the two official archives was unavoidable. The full research trail for this section lives in `DATA_ACCESS.md`; this section summarizes the outcome and the reasoning.

3.1 Metadata first, imagery last

The two official metadata files (~4.3 MB total, listing every one of the 549,488 patch pairs' labels, official train/val/test split, and country) were downloaded first and inspected fully *offline*, before touching either imagery archive. This alone resolved: the class list (19 official CORINE-derived classes), confirmation that Sentinel-1/Sentinel-2 pairing is a lossless 1:1 lookup by patch ID (verified directly — zero nulls, zero duplicates across all 549,488 rows), and that the official split is geographically de-correlated by design (the reBEN paper's own stated contribution) — meaning it should be used as-is rather than re-split, to avoid reintroducing the spatial leakage the dataset authors specifically engineered against.

3.2 A cheap, bounded probe before any large transfer

A small HTTP range request (30 MB from the start of each archive) was used to inspect the archives' internal structure before committing to any download strategy. This confirmed two things that shaped everything downstream:

- Both archives are **single, non-seekable zstd compression frames** — there is no way to jump to an arbitrary point in the archive; the only option is to decompress sequentially from byte zero.
- Archive entries are ordered **alphabetically by product name**. For Sentinel-2, product names start with acquisition date, so this is effectively chronological order. For Sentinel-1, product names start with the satellite ID (S1A vs. S1B), producing two large contiguous blocks rather than a date order.

This meant a subset could be obtained by **streaming from byte zero and writing only the matching patches to disk, aborting the connection once past the last needed product** — without ever downloading the full archive. The practical question became: which products to select so that this stream stays short.

3.3 Subset selection: five tiles, one satellite pass, 4,200 patches

Working entirely from the already-downloaded metadata:

1. The five alphabetically-earliest Sentinel-2 products were selected (Austria, Finland ×2, Ireland, Portugal) — cheap to reach since the S2 archive is itself alphabetically ordered. These five products

alone yield 34,190 patches covering **all 19** official classes.

2. Patches were further restricted to those whose paired Sentinel-1 scene comes from the **S1A** satellite (not S1B) — since the S1 archive is split into a contiguous S1A block followed by S1B, this single filter cut the required S1 stream from an estimated ~35 GB down to ~1.4 GB, while still preserving all 19 classes and all three splits. This left **13,932 candidate paired patches**.
3. A stratified sample of **4,200 patches** was drawn from that candidate pool, proportional to the official train/validation/test split sizes, sized to train comfortably on CPU-only hardware within the exercise's time budget.

> **Result: ~1.8 GB downloaded in total** (4.3 MB metadata + 1.3 GB raw GeoTIFF imagery for 4,200 paired patches), against a 118 GB full-archive alternative — a ~65× reduction, achieved by exploiting known archive structure rather than guessing. (Section 9 revisits this construction with a different class-selection strategy, drawing a larger sample from the same 5-tile pool at the cost of a second sequential pass over the archives.)

A caveat disclosed openly rather than hidden: restricting to S1A-paired patches is a convenience constraint for this exercise's bandwidth/time budget, not a scientific one — it slightly biases the sample toward whichever sub-area of each tile happened to be nearer an S1A overpass. This is a reasonable trade-off at this scale, and is stated plainly as a sampling caveat in the README rather than presented as a fully representative geographic sample.

4. Simulated Cloud Masking

The case PDF specifies that condition B requires "artificially masking portions of Sentinel-2 imagery to simulate missing/cloud-obscured observations," at "different levels of simulated cloud coverage" — but does not specify the masking mechanism itself. This was treated as an open design decision, documented rather than silently assumed.

Implementation: 1–3 randomly placed, randomly sized rectangular occlusions per patch, with combined area iteratively adjusted against the actual accumulated mask (not a naive pre-computed area budget, which was found — and fixed — to systematically under-shoot the target coverage due to rectangle overlap) until the target coverage fraction is reached to within about 1% average error. Masked pixels are replaced with that band's mean value over the *unmasked* region of the same patch — a neutral "no information" fill, rather than zero, which would otherwise read as a real (and highly informative) extreme reflectance value. Four coverage levels were evaluated throughout: **0%, 25%, 50%, and 75%**.

> This is a deliberate simplification of real cloud shape, transparency, and atmospheric correlation — documented as a design choice for speed and reproducibility, not claimed to be a realistic cloud simulator. Testing against BigEarthNet's own naturally cloud/shadow-flagged patches (available in the metadata, excluded from this subset) is listed as a natural follow-up in Section 9.

5. Model Design

5.1 The minimum baseline (conditions A and B)

A small CNN — four convolutional blocks (32→64→128→256 channels, each with batch normalization, ReLU, and 2×2 max-pooling), followed by global average pooling and a linear classifier — trained from scratch on clean Sentinel-2 imagery (12 bands, resized to 120×120 pixels), multi-label BCE loss (a single patch can carry several land-cover labels simultaneously in BigEarthNet's native scheme). The *same trained model* is then re-evaluated on masked inputs at each coverage level — exactly the "train once, evaluate under degradation" recipe the case specifies as the floor. This is intentionally not state-of-the-art: about 400,000 parameters, no pretrained weights, ~70–80 seconds per training epoch on a 12-core CPU.

5.2 The SAR-assisted contribution (condition C)

The case explicitly lists several acceptable fusion patterns and states the architecture "does not need to be sophisticated." **Round 1** implemented the simplest of these: early concatenation of the 12 optical bands with the 2 SAR bands (VV, VH polarization) into a single 14-channel input to the same CNN backbone. **Round 2** added two more variants to actually test open questions rather than assume answers:

- **C-late** — a genuinely different architecture: separate encoder towers for optical and SAR, each pooled to its own feature vector, concatenated *after* encoding rather than before ("separate encoders + feature fusion," the case's other listed option).
- **C-fixed50** — identical architecture to the round-1 model, but trained at a single fixed 50% mask coverage instead of a randomized 0–75% range, to test whether training-time coverage diversity actually helps generalization.

SAR backscatter values (linear power/amplitude) were converted to decibels (`10*log10(x)`) before normalization, a standard preprocessing step for SAR data that produces a more Gaussian-like, numerically stable distribution than raw linear values.

6. Round 1 — Baseline Results

All numbers below are on the held-out test split (1,063 patches), 8 training epochs, no hyperparameter tuning, evaluated with the naive all-19-class macro-F1 (round 2, Section 7, revises this metric after finding a validity issue — read this section as the initial, direct answer).

Condition	Coverage	Macro F1	Micro F1	Hamming Acc.
A — optical only, clean	0%	**0.407**	0.618	0.896
C — SAR-assisted	0%	0.389	0.602	0.893
B — degraded optical only	25%	0.333	0.564	0.892
C — degraded optical + SAR	25%	**0.384**	0.605	0.892
B — degraded optical only	50%	0.240	0.497	0.882
C — degraded optical + SAR	50%	**0.386**	0.604	0.890
B — degraded optical only	75%	0.207	0.468	0.872
C — degraded optical + SAR	75%	**0.385**	0.598	0.887

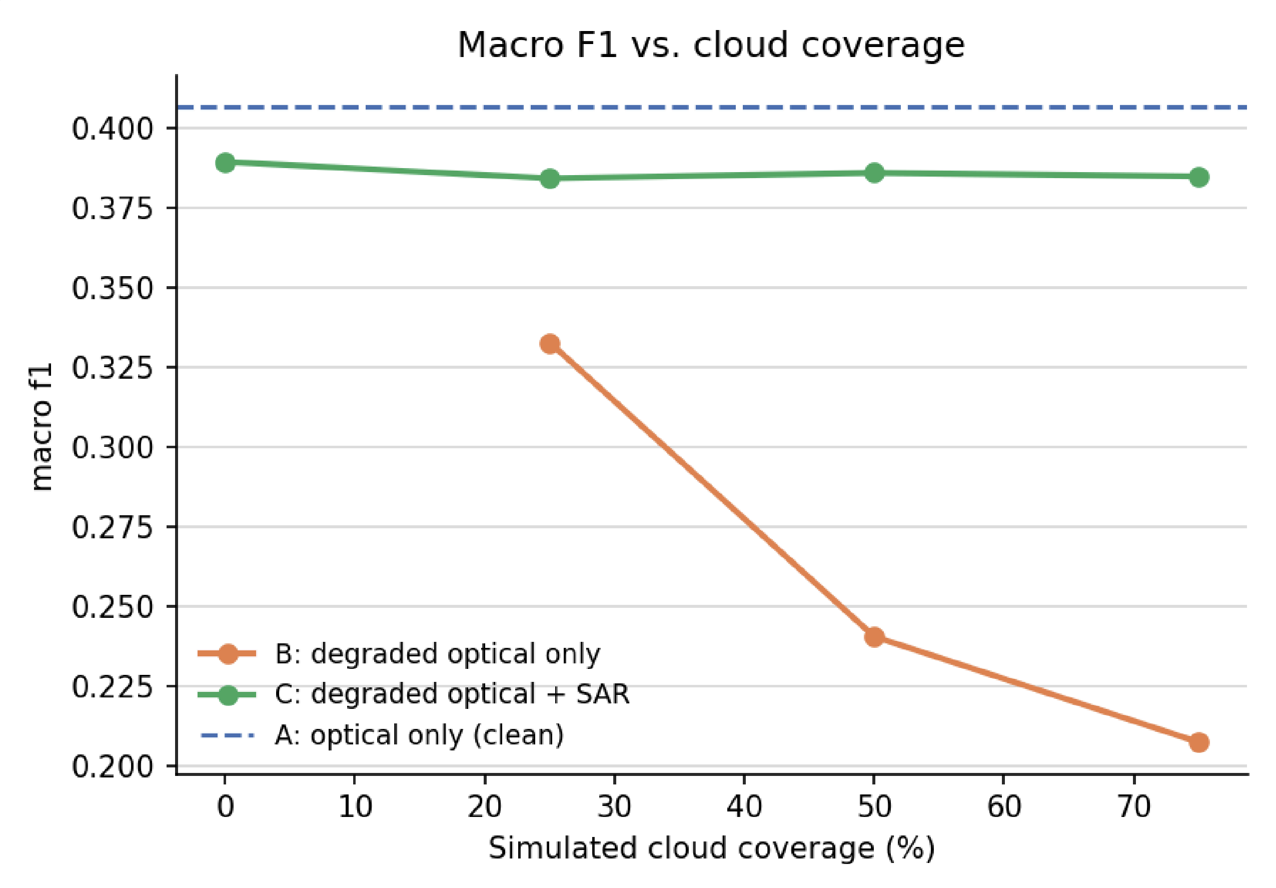

Figure 1. Round-1 macro-F1 vs. simulated cloud coverage. Optical-only (B) collapses as coverage increases; the SAR-assisted model (C) stays nearly flat.

The headline round-1 finding: as cloud coverage increases, optical-only performance **drops 49%** (0.407 → 0.207), while the SAR-assisted model stays essentially flat (0.389 → 0.385, about a 1% relative change) — matching the case's stated motivation directly. One honest asymmetry noted even at this stage: at 0% coverage, the clean optical model (0.407) still edges out the SAR-assisted model (0.389), because the SAR model is trained with random mask augmentation and never sees purely clean data at full weight — a reasonable robustness/peak-accuracy trade-off, not a bug.

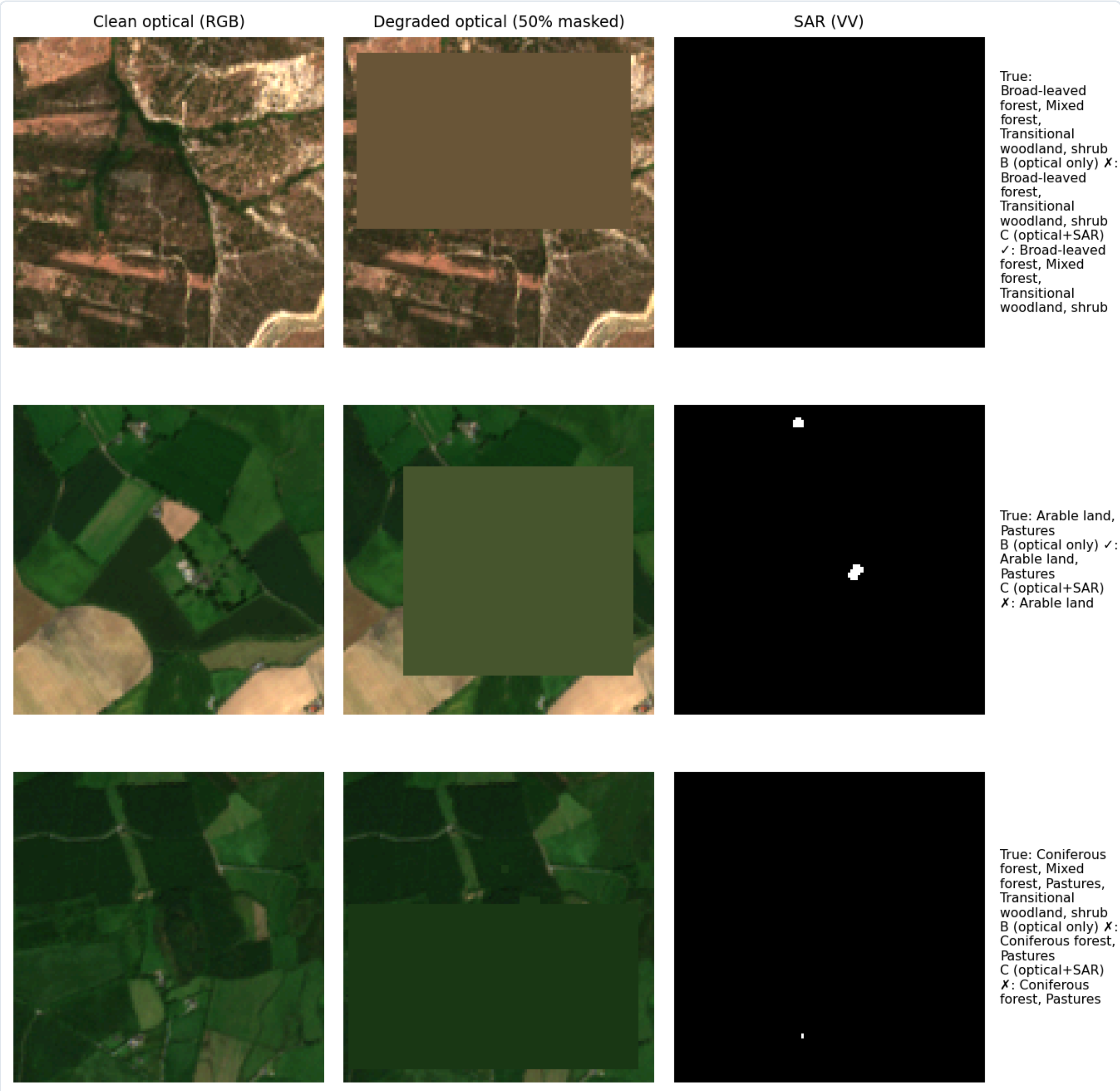

Figure 2. Three representative test patches at 50% coverage: a case where SAR corrects an optical-only error, a case where SAR introduces one, and a case where both fail. Shown deliberately as a mix, not cherry-picked wins.

An uncertainty finding from this round, using mean predictive entropy as a simple proxy: the optical-only model's entropy stayed essentially flat even as its accuracy collapsed by half — it was failing *confidently*, not appropriately uncertainly, which is the worse failure mode for any system that might use predicted confidence to decide when to trust a result.

7. Round 2 — Finding a Bug, and Going Deeper

Round 1 answered the core question directly. Round 2 went back and asked whether the evaluation itself was trustworthy, rather than immediately building more on top of it.

7.1 A real bug: three classes were unlearnable or unscoreable by construction

Investigating *why* several classes scored zero F1 in round 1 (rather than attributing it to "small subset, no tuning time" and moving on) revealed a genuine data-construction issue:

Class	Train examples	Validation examples	Test examples
Beaches, dunes, sands	**0**	3	30
Marine waters	51	**0**	**0**
Coastal wetlands	29	**0**	**0**

"Beaches, dunes, sands" has zero training examples anywhere in the 5-tile candidate pool — no model could ever learn it, regardless of loss weighting or training time. "Marine waters" and "Coastal wetlands" have zero validation/test examples — they may well have been learned, but could never be scored (scikit-learn's F1 silently returns 0 for a class with no ground-truth positives, which reads as a failure but is actually a missing-data artifact). This was confirmed to be a genuine geographic side effect of the 5-tile selection (BigEarthNet's official split is region-based, so a class confined to one small area can land entirely in one split) — **not** a sampling-code bug: every class that has examples available in a split does appear in that split's final subset; these three simply have none to begin with.

> **The fix:** a new `macro_f1_valid` metric was added, computed only over the 16 classes with at least one example in all three splits, reported alongside the original all-19-class metric for direct comparability. This is exactly the kind of correction the case's own "is your evaluation scientifically valid" criterion is asking for.

7.2 Class-balanced loss

Inverse-frequency positive-class weighting was added to the training loss (computed from train-split label frequency only, capped to avoid instability from near-empty classes).

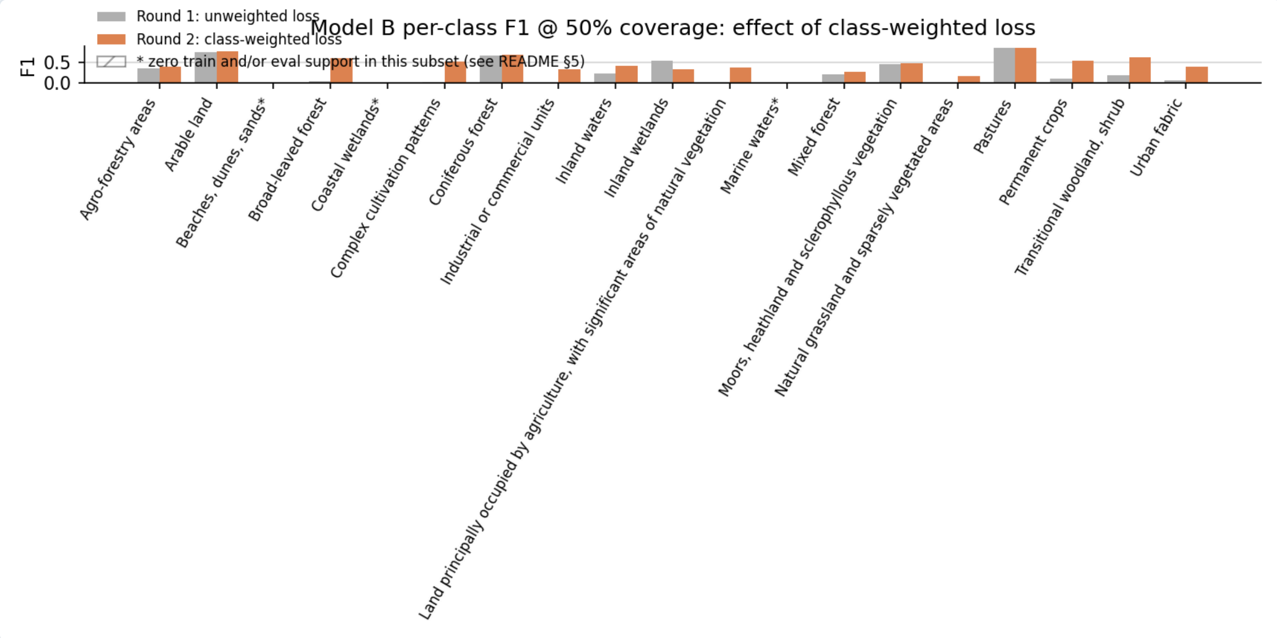

*Figure 3. Effect of class-balanced loss on the optical-only model's per-class F1 at 50% coverage. Genuinely rare-but-learnable classes (e.g. "Industrial or commercial units," "Complex cultivation patterns") improve; the three structurally-excluded classes (marked with *) predictably stay flat, exactly as the Section 7.1 diagnosis predicts.*

7.3 Three SAR-assisted variants, tested rather than assumed

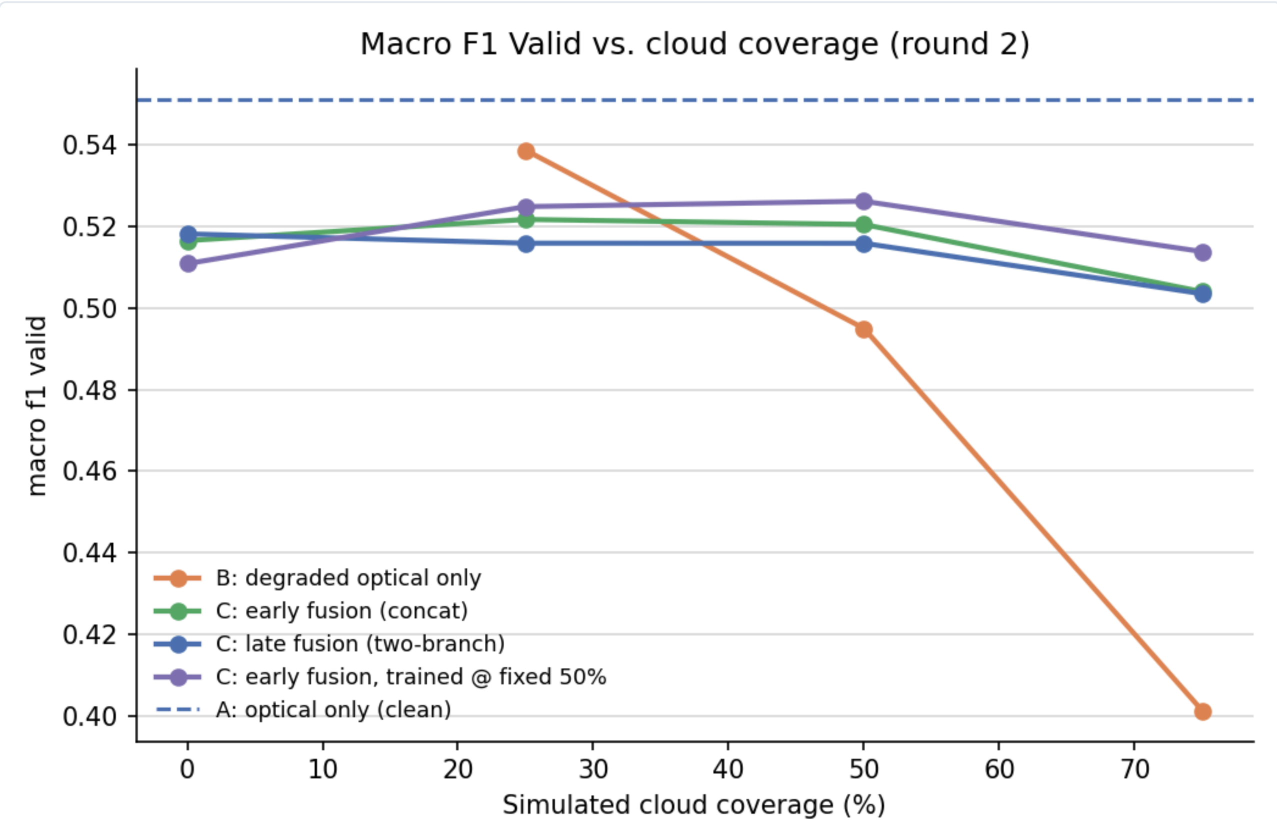

Figure 4. Round-2 corrected macro-F1 (16 valid classes, class-balanced loss) across all four model variants.

Coverage	B	C-early	C-late	C-fixed50

0%	**0.551**	0.516	0.518	0.511
25%	**0.539**	0.522	0.516	0.525
50%	0.495	0.520	0.516	**0.526**
75%	0.401	0.504	0.503	**0.514**

Finding 1 — the crossover point moved. With class-weighted loss, the optical-only baseline is much stronger than round 1 showed, and holds its own up to 25% coverage. The SAR crossover point shifts from ~25% (round 1) to ~50% coverage. The qualitative conclusion survives — SAR clearly wins under heavy degradation, a ~28% relative gap at 75% coverage — but the exact crossover point was partly an artifact of an under-trained baseline, not a fixed property of the task. This revision is reported directly rather than only keeping round 1's more dramatic number.

Finding 2 — more architectural complexity did not help. The two-branch late-fusion model (C-late) took about 55% longer per training epoch than the simple concatenation model and produced statistically indistinguishable results — direct empirical support for the case's own claim that the architecture "does not need to be sophisticated."

Finding 3 — a mildly counterintuitive training-regime result. A model trained at one *fixed* 50% coverage generalized better across the full 0–75% range than one trained on *randomized* 0–75% coverage — worst at 0% coverage (having never seen clean data), but best everywhere from 25% onward. This is reported as a genuine finding, selected by an automatic rule (highest mean score under degradation) rather than a manually cherry-picked result.

7.4 Calibration: temperature scaling and Expected Calibration Error

Beyond round 1's raw-entropy proxy, round 2 adds a standard, quantifiable calibration metric: a single temperature parameter fit on validation-set logits (never test data), then Expected Calibration Error (ECE) measured before and after, at 50% coverage.

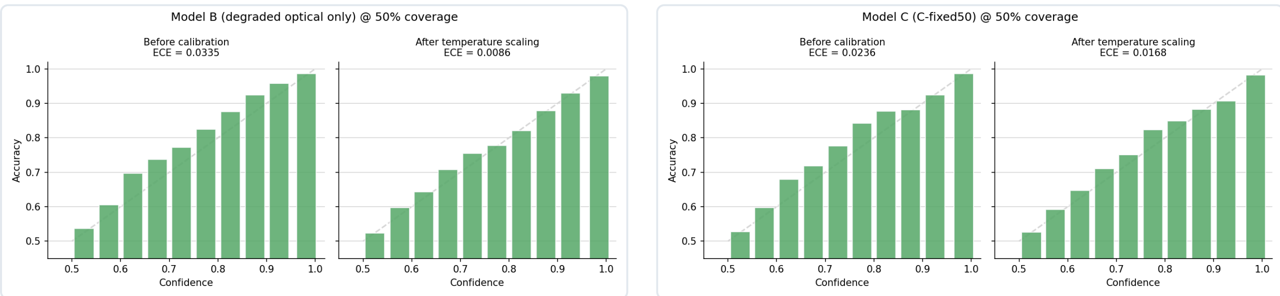

Figure 5. Reliability diagrams before/after temperature scaling: model B (left, ECE 0.0335→0.0086) and the best round-2 SAR model (right, ECE 0.0236→0.0168).

Both models were found to be mildly *under*confident at this operating point (not overconfident) — a nuance that refines, rather than contradicts, round 1's separate finding that B's entropy barely tracks increasing coverage severity. Both things are true simultaneously: B's probabilities are reasonably well-calibrated in an absolute sense at one fixed coverage level, while its *sensitivity* of confidence to input difficulty across coverage levels remains poor.

8. Class-Confusion Analysis

The case's own uncertainty-reasoning criterion names this explicitly: "which classes confuse each other." Per-class F1 alone doesn't answer this — it shows a class was missed, not what the model said instead. Because this is a multi-label problem, a standard single-label confusion matrix doesn't directly apply; instead, for every test sample and every true class the model misses, every class it wrongly adds on that same sample is counted as a "confused-for" pair.

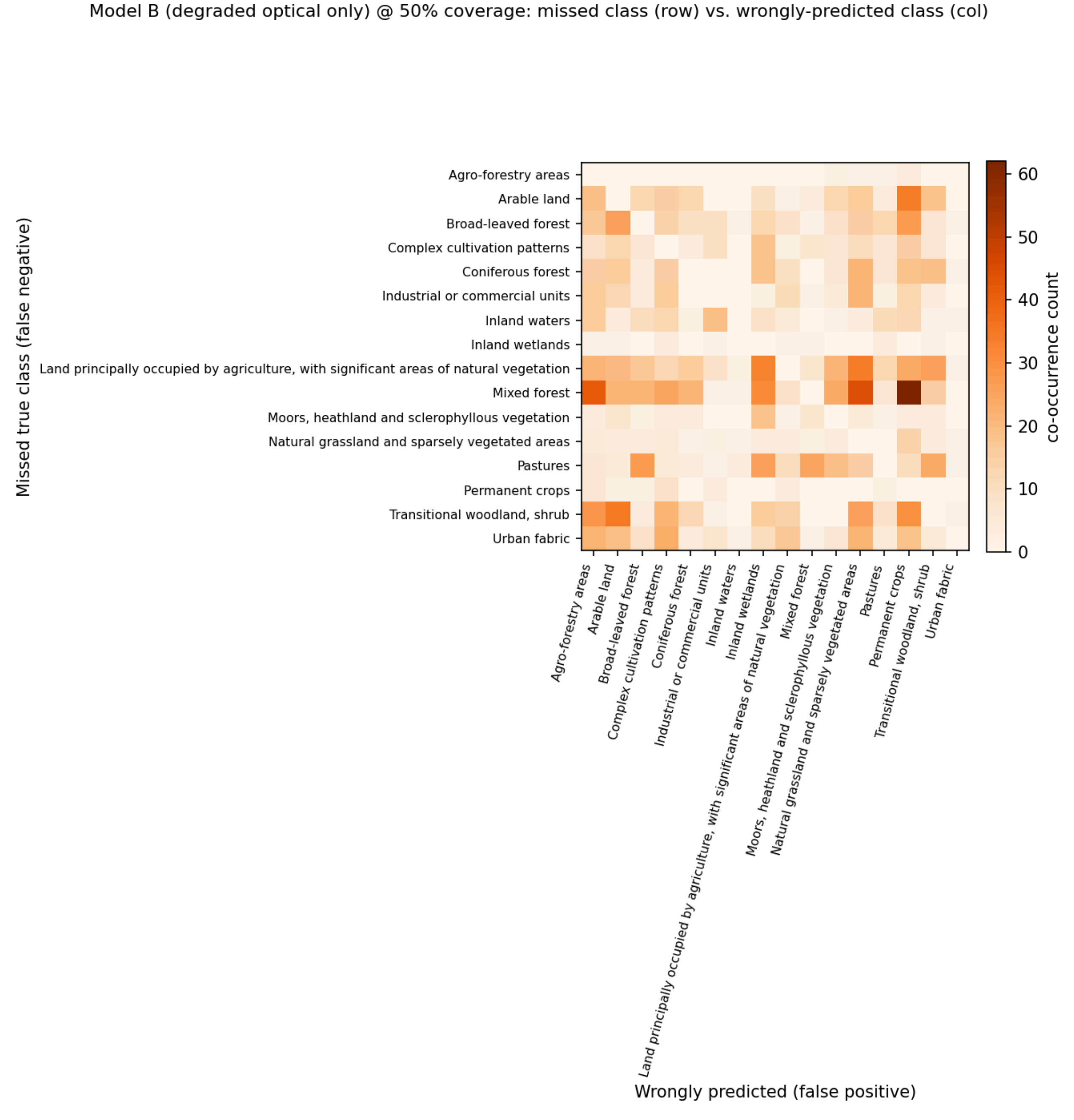

Figure 6a. Model B (optical only) confusion pattern at 50% coverage.

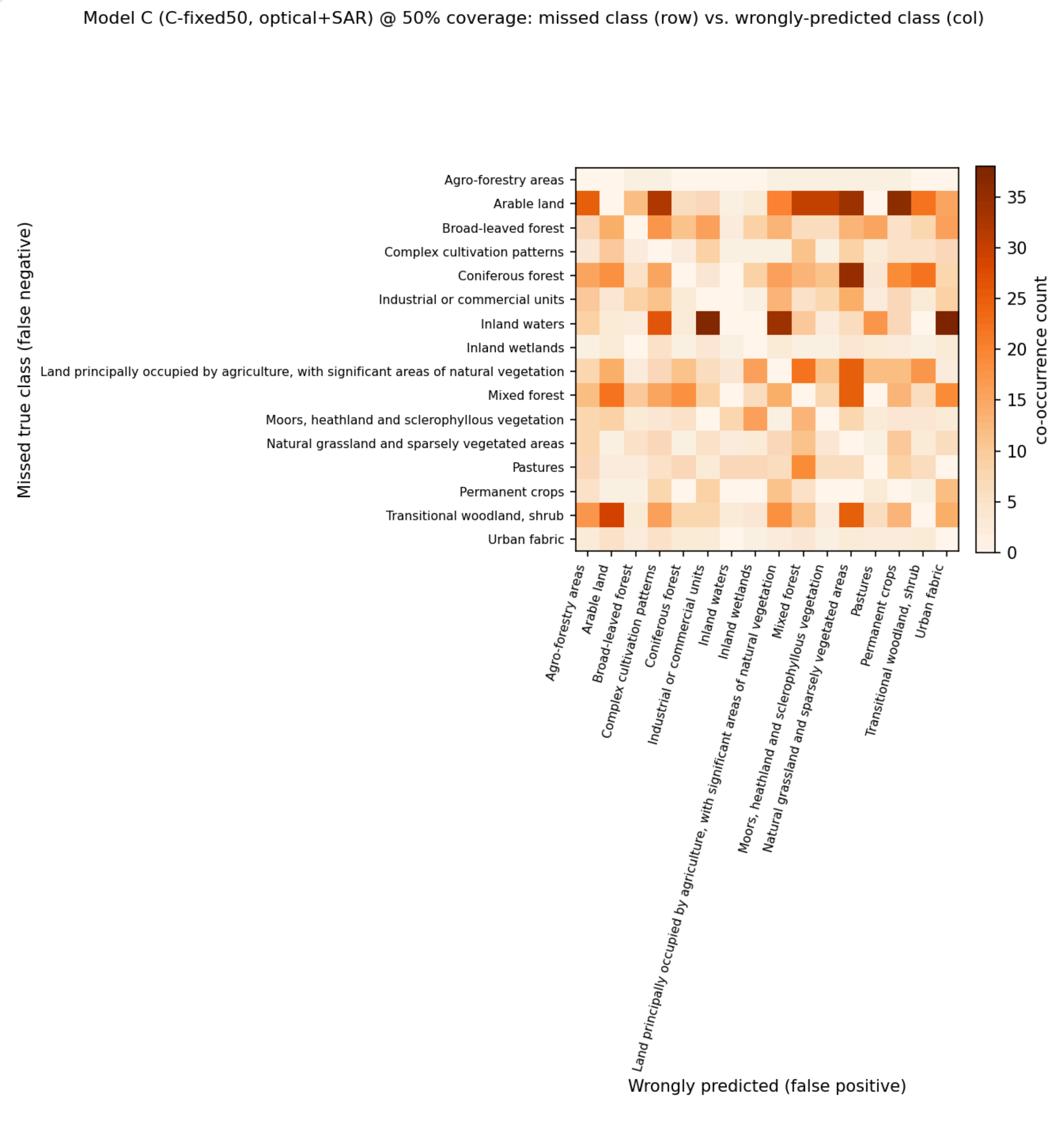

Figure 6b. Model C (C-fixed50, optical+SAR) confusion pattern at 50% coverage.

Model B's dominant confusion is missing "Mixed forest" and instead predicting "Permanent crops," "Natural grassland," or "Agro-forestry areas" — semantically quite different land covers, consistent with the model falling back on whatever texture remains once enough of a forest patch is masked, rather than degrading toward a visually similar forest type.

Model C's confusion pattern is different in kind, not just smaller: its top confusion is missing "Inland waters" and predicting "Urban fabric" or "Industrial or commercial units" — a surprising water-vs-built-up mix-up with no obvious optical explanation. A plausible mechanism: calm inland water and certain urban/industrial surfaces can produce superficially similar low-texture SAR backscatter, and the fusion model may occasionally import this confusion rather than resolve it. This is a genuine, non-obvious failure mode that aggregate F1 numbers alone would never surface.

9. Round 3 — Fewer Classes, More Examples Per Class

Rounds 1–2 stratified a fixed patch budget across all 19 classes present in the 5 tiles, including several with only 29–51 total examples in the entire 13,932-patch candidate pool. In a multi-label dataset, "examples per class" is not fixed by total patch count — it is set by which classes are chosen to model and how a fixed budget is allocated across them. Round 3 asks directly: would trading class breadth for per-class depth do better?

9.1 Design

The 8 best-supported classes were selected — verified to have solid examples in **all three** official splits (Pastures, Coniferous/Mixed/Broad-leaved forest, Arable land, Transitional woodland/shrub, Complex cultivation patterns, and the mixed-agriculture class) — using train-split frequency (cross-checked against all-split frequency: identical ranking either way; see Section 10.3). These 8 classes already appear in **97.8%** of the entire candidate pool, so restricting to them costs almost no data. A much larger sample — **8,000 patches** (vs. 4,200) — was drawn from the same already-scanned 5-tile candidate pool: no new tile selection, no new geographic scope, just a second sequential pass over the same byte range (unavoidable given the archives' non-seekable structure, Section 3.2).

	Round 1/2 (19 classes, 4,200 patches)	Round 3 (8 classes, 8,000 patches)
Weakest kept class, train examples	~250–290	**830+**
"Pastures," train examples	859	**1,676**
Classes with 0 support in some split	3 (worked around)	**0** by construction

9.2 Results

Same architecture and training recipe as round 2 (class-balanced loss, 10 epochs), using round 2's winning training regime for the SAR model (concatenation fusion, fixed 50% training coverage). Evaluated on round 3's own held-out test set (2,049 patches):

Coverage	B (optical only)	C (optical + SAR)
0%	**0.672**	0.644
25%	0.656	0.658
50%	0.611	**0.666**
75%	0.527	**0.662**

The qualitative pattern matches rounds 1–2 (B collapses under heavy degradation, C stays flat), and absolute numbers are substantially higher — but that is expected and not, by itself, evidence that "more depth helps": an 8-class task restricted to common, visually distinct land covers is simply an easier task, independent of training data volume. Isolating the effect of depth requires holding classes and test set fixed and varying only the training data.

9.3 Does more depth actually help? A same-classes, same-test-set comparison

Round 2's already-trained 19-class models were re-evaluated on round 3's test set, predictions sliced down to the same 8 classes — an apples-to-apples comparison of shallow training data (round 2, ~250–860 examples/class) vs. deep training data (round 3, ~830–1,700+ examples/class), same classes, same test patches, same evaluation code.

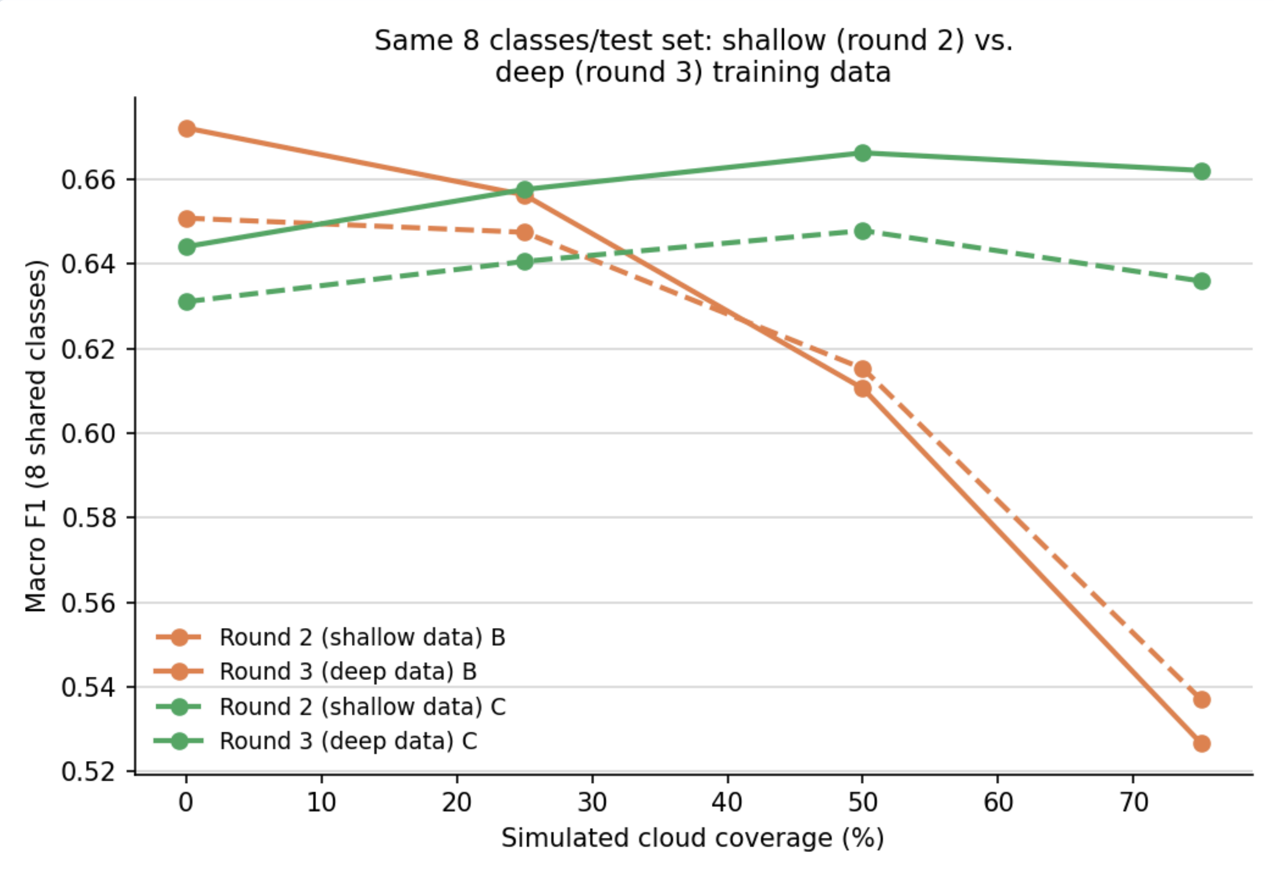

Figure 7. Same 8 classes, same test set: shallow (round 2) vs. deep (round 3) training data, model B and model C.

Coverage	Shallow B	Deep B	Shallow C	Deep C
0%	0.651	0.672 (+0.021)	0.631	0.644 (+0.013)
25%	0.647	0.656 (+0.009)	0.641	0.658 (+0.017)
50%	0.615	0.611 (−0.005)	0.648	0.666 (+0.018)
75%	0.537	0.527 (−0.010)	0.636	0.662 (+0.026)

The answer is genuinely mixed, not a clean "yes." For the SAR-assisted model, more depth helped consistently, and by an *increasing* margin as coverage increases — the smallest gain at 0% coverage, the largest at 75%. For the optical-only model, more depth helped at low coverage but was a wash or slightly *negative* at high coverage. A plausible explanation: under heavy masking, B's bottleneck is not "not enough training examples" — it is "not enough visual signal left in the input, period." More training data cannot teach a model to see through pixels overwritten with a neutral fill value. This is a genuine, non-obvious finding: **depth and SAR-fusion solve different problems** — depth alone would not have closed the B-vs-C gap that SAR closes.

9.4 Honest costs

Because the archives are single non-seekable zstd frames, round 3's larger sample required a second full sequential pass over the same byte range rounds 1–2 already scanned — there is no way to resume or append to a prior partial extraction. Total network transfer across both extraction passes: **~3.6 GB** (vs. ~1.8 GB if this design had been chosen from the start). Total on-disk footprint (raw + cache, excluded from version control, fully reproducible): **~10.7 GB**. Dropping to 8 classes also means no more coastal, marine, industrial, or urban land covers in the story — a real reduction in scope, not a free win, though arguably closer to the case's literal instruction to "select several land-cover classes" than rounds 1–2's approach of keeping whatever happened to be present in the chosen tiles.

10. Methodological Audit: Checking for Bias and Leakage

Requested explicitly as a check on this project's own validity, not a routine step. This section reports what was checked, what was found and corrected, and what was checked and found to be fine — all of it, not only the reassuring parts.

10.1 Issue found and corrected: a test-set-informed model selection in round 2

Round 2 trained three SAR-fusion variants and picked "the best one" by comparing their mean macro-F1 **on the test set**. This is a real methodological issue: comparing several trained candidates against test-set performance and reporting the winner as "the" result is a mild form of test-set-informed selection — the reported number for the winner is optimistically biased relative to what a single, a-priori-chosen architecture would show, because picking the best of several noisy estimates tends to overstate how good that estimate is.

Correction: the selection was re-derived using each variant's best **validation-set** score achieved during training — the same quantity already used for per-epoch checkpoint selection, compared across variants instead of only within one.

Variant	Best validation macro-F1 (valid)	Test-set mean macro-F1 (coverage > 0)
C-early	0.5758	0.5153
C-late	0.5674	0.5117
C-fixed50	**0.5759**	**0.5215**

> **C-fixed50 is still the nominal winner** — the qualitative conclusion in Section 7 stands. But the validation-based margin (0.5759 vs. 0.5758 — a difference of 0.0001) is far smaller than the test-set margin made it look, and is almost certainly within run-to-run noise for a single seed. The honest summary: C-early and C-fixed50 are statistically indistinguishable on this evidence; C-late is the one variant that looks consistently worse on both validation and test. This is the single most important thing to weigh when reading this project's specific numeric claims about which fusion regime is "best."

10.2 Checked and confirmed clean: no cross-round data leakage

Round 3 draws a much larger sample from the same candidate pool rounds 1–2 used. Since round 2's model (trained on round 1/2's train split) is evaluated on round 3's test set in Section 9.3, any overlap between round 2's training patches and round 3's test patches would bias that comparison in round 2's favor. Checked directly: **zero patch-ID overlap** between round 2's 2,040 training patches and round 3's 2,049 test patches (and zero overlap in every other cross-split pairing checked). This is guaranteed by construction — both rounds draw from the same official, never-reassigned split column — but was verified empirically rather than only argued logically.

10.3 Checked and confirmed harmless: class selection did not depend on test-set data

Round 3's 8 target classes were selected by ranking frequency across the entire candidate pool (all three splits combined) — technically giving the selection process access to test-split label composition. Re-ranked using train-split frequency only: the identical 8 classes come out on top, in the same order. Because these classes are overwhelmingly common in every split (not narrowly enriched in test specifically), the all-split shortcut happened to be harmless here — but this was checked, not assumed, and the distinction matters in general: a class selection that did depend on test-specific frequency patterns would have been a genuine leakage concern.

10.4 Other things checked

- **Normalization statistics** and **class weights** — both computed from train-split data only, confirmed by reading the preprocessing and weighting code directly.
- **Temperature scaling** — fit on validation-set logits only; test-set logits used only for the final before/after measurement, never for fitting the temperature parameter.
- **Checkpoint selection during training** — always uses the validation split's score, never test.
- **Qualitative success/failure case selection** — sampled via a seeded random choice from the relevant outcome pool, not hand-picked for narrative effect.
- **No hyperparameters** (learning rate, epochs, batch size, architecture width) were iterated based on observed test-set performance — all were fixed a priori per run.

This was a targeted check of the specific mechanisms in this codebase, not an exhaustive statistical audit. It does not — and cannot, without the multi-seed runs listed in Section 11 — rule out run-to-run noise being a larger contributor to the reported gaps than the narrative attributes to any single cause. That limitation is structural, not something this audit could check its way out of.

11. Honest Limitations and What's Next

Updated after rounds 2–3 and the audit in Section 10, in rough priority order:

1. **Multi-seed runs with confidence intervals** — promoted to the top item after Section 10.1's finding. Every result in this document, across all three rounds, is a single train/eval run, and Section 10.1 showed directly that a test-set-based comparison can make a razor-thin (0.0001) validation margin look like a clear (0.006) test-set win. 3–5 seeds per condition would let every table in this document report real error bars instead of point estimates.
2. **A larger, more geographically diverse subset.** This project used 5 tiles from 4 countries and an S1A-only pairing filter for bandwidth/time reasons throughout every round. A broader subset would let "Beaches, dunes, sands" actually become learnable, and would let a future round test whether round 3's "8 classes, deep data" design generalizes beyond this geographically narrow sample.
3. **More realistic cloud simulation**, tested against BigEarthNet's own naturally-flagged cloud/shadow patches rather than synthetic rectangles.
4. **Calibration analysis across all four coverage levels** (this project only checked 50%), to see whether calibration degrades with coverage the way accuracy does.
5. **A closer look at the C-fixed50 training-regime result** — given Section 10.1 found its margin over C-early is likely noise, this is now lower priority; if pursued, try more fixed training-coverage values with multiple seeds each.
6. **Extend round 3's depth-vs-breadth question to a middle ground** — e.g. 12–14 classes at an intermediate depth — to map the trade-off curve more finely than the two endpoints tested so far.

> None of these gaps were hidden or discovered by omission — each is a deliberate scope decision made under the case's own explicit constraint not to over-invest in tuning or scale beyond what a 24–48 hour exercise calls for. The choice to disclose them plainly, rather than present a falsely complete picture, is itself part of the "scientifically valid evaluation" and "reasons about uncertainty" criteria this exercise asks for.

12. Project Artifacts and Reproduction

12.1 External resources used

- **Dataset:** BigEarthNet v2.0 / reBEN (Clasen et al., 2024), Zenodo record 10891137, CDLA-Permissive-1.0 license.
- **Libraries:** PyTorch & torchvision, NumPy/pandas/pyarrow, scikit-learn, tifffile + Pillow, zstandard, matplotlib, requests.
- No pretrained model weights were used — every model was trained from scratch, since a 12/14-channel multispectral input does not align with standard 3-channel ImageNet-pretrained backbones without extra modification.
- Rounds 2 and 3 introduce no new external dependencies beyond the above (PyTorch's built-in LBFGS optimizer is used for temperature-scaling calibration).

12.2 Reproducing this project end-to-end

```
pip install -r requirements.txt
python src/download_metadata.py           # ~4.3 MB
python src/select_subset.py               # offline metadata filtering
python src/extract_subset.py              # streams ~1.8 GB from Zenodo
python src/preprocess.py                  # builds cached tensors + norm stats
python src/run_experiment.py              # round 1: trains A & C, evaluates across coverage
python src/run_experiment_v2.py           # round 2: weighted loss, 3 C variants, calibration
python src/confusion_analysis.py          # round 2: class-confusion matrices (no retraining)

python src/select_subset_v3.py                       # round 3: 8-class selection
python src/extract_subset.py subset_patches_v3.csv   # round 3: 2nd pass, same tiles, ~1.8 GB
python src/preprocess.py subset_patches_v3.csv _v3   # round 3: cache + norm stats
python src/run_experiment_v3.py                      # round 3: trains B & C, cross-round comparison
```

12.3 Repository layout

```
src/                    all pipeline code, one script per pipeline stage
data/                   metadata, subset selection CSVs (rounds 1/2 + round 3's _v3 variants),
                        raw extracted GeoTIFFs (excluded from version control, ~2.75 GB)
outputs/cache/          preprocessed tensors, shared across all rounds (excluded, ~7.9 GB)
outputs/checkpoints/    trained model weights (rounds 1-3, 9 models total)
outputs/metrics/        results tables, per-epoch training history, calibration summaries,
                        cross-round comparison data
outputs/figures/        every figure referenced in this document and in README.md
README.md               top-level narrative and results (companion to this document)
REQUIREMENTS.md         full requirements-extraction pass from the case study PDF
DATA_ACCESS.md          full data-access research trail (Section 3 of this document, in detail)
```